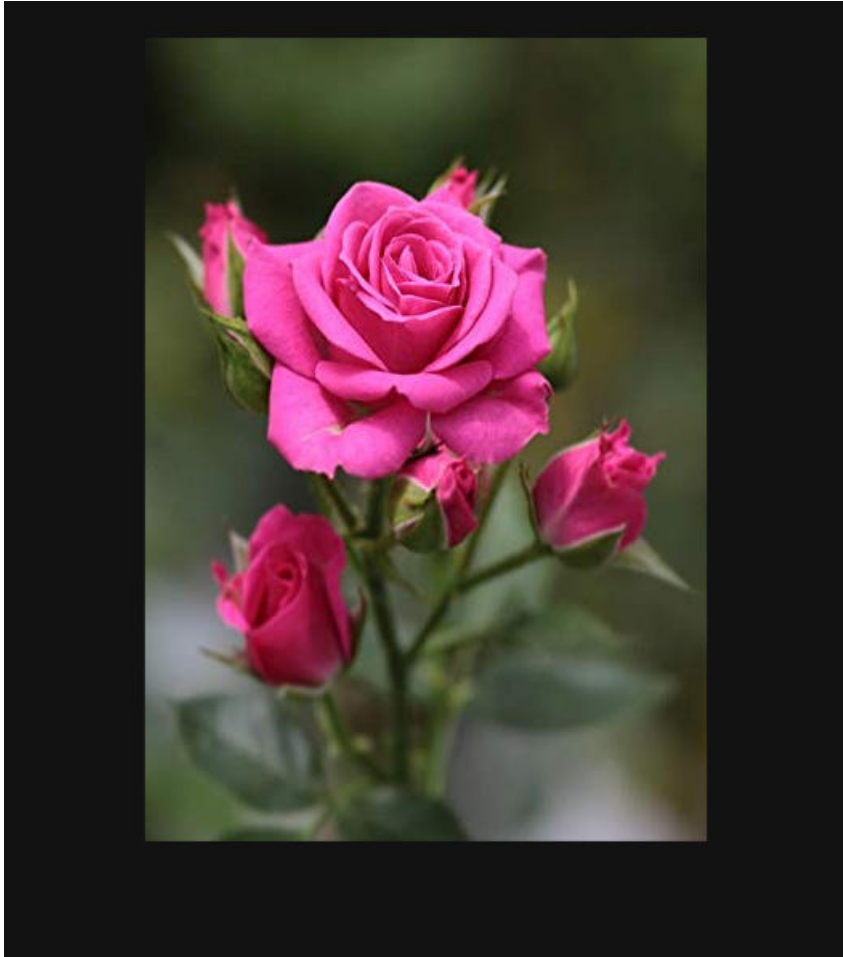


```
% TASK 1:
% Load the image and convert pixel values to a double-precision range of [0, 1]
[cite: 91, 93, 99]
I = imread('C:\Users\Administrator\OneDrive - Habib University\Sem 4\Signals &
Systems\MatLab\Lab07\flower.jpg');
I = im2double(I);
figure, imshow(I)
```



```
whos I
```

Name	Size	Bytes	Class	Attributes
I	500x351x3	4212000	double	

```
[R, G, B] = imsplit(I);
whos R
```

Name	Size	Bytes	Class	Attributes
R	500x351	1404000	double	

```
W = 5;
h = ones(W) ./ W.^2;
```

```
% Apply 2D convolution to each color plane to blur
```

```
Ravg = conv2(R, h, "same");
```

```
Gavg = conv2(G, h, "same");
```

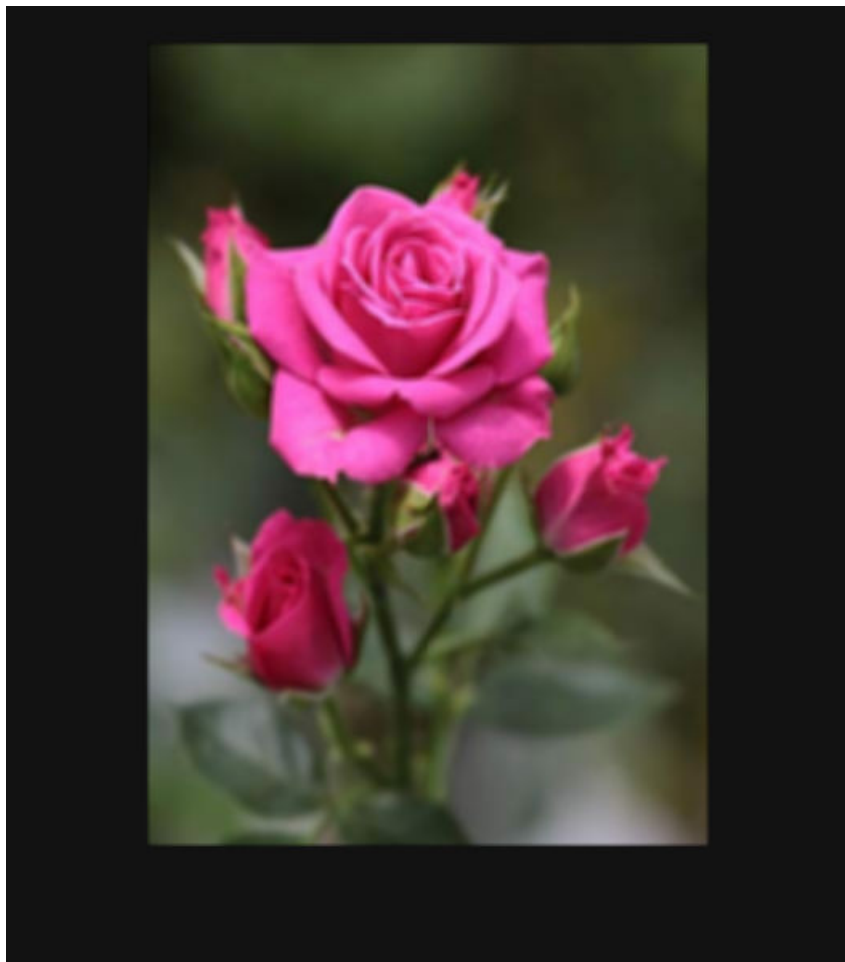
```
Bavg = conv2(B, h, "same");
```

```
Iavg = cat(3, Ravg, Gavg, Bavg);
```

```
whos Iavg
```

Name	Size	Bytes	Class	Attributes
Iavg	500x351x3	4212000	double	

```
figure, imshow(Iavg, [])
```



```
%% Task 2:
```

```
n = 0:50;
```

```
% Creating the input signal
```

```
x = double((sin(0.5*n) + 0.2) < 0);
```

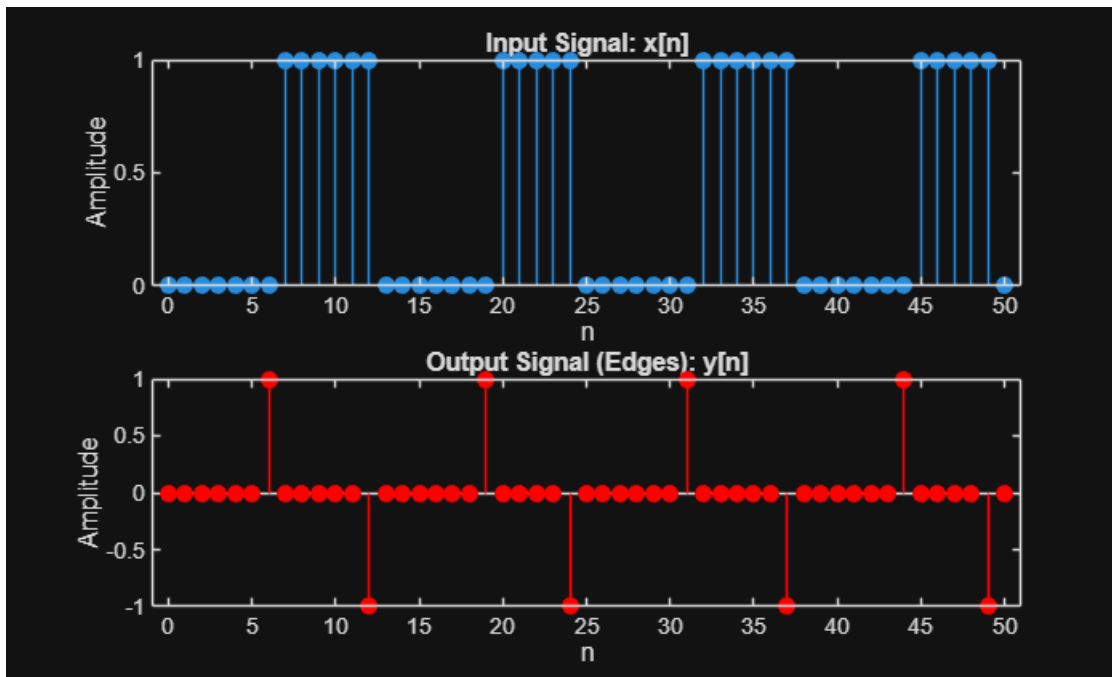
```
% Defining the first difference filter impulse response
```

```
h = [1, -1];
```

```
y = conv(x, h, 'same');
```

```
figure;
subplot(2,1,1);
stem(n, x, 'filled');
title('Input Signal: x[n]');
xlabel('n'); ylabel('Amplitude');

subplot(2,1,2);
stem(n, y, 'filled', 'r');
title('Output Signal (Edges): y[n]');
xlabel('n'); ylabel('Amplitude');
```



The first-difference filter, defined by the impulse response, acts as a basic edge detector. By comparing the current value of the signal to its previous value, the filter identifies transitions in the input.

Positive Spikes: A value of +1 in the output indicates a transition from 0 to 1.

Negative Spikes: A value of -1 in the output indicates a transition from 1 to 0.

Zero Values: When the signal is constant (all 0 or all 1s), the output is 0 because there is no change between adjacent samples.

```
% Task 3: Edge Detection on Image Data
```

```
% Load the echart matrix
```

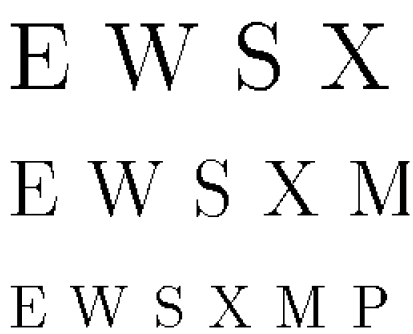
```
load('C:\Users\Administrator\OneDrive - Habib University\Sem 4\Signals & Systems\MatLab\Lab07\echart.mat');
```

```
% Display the original image
```

```
figure;
imshow(echart);
```

```
title('Original E-Chart Image');
```

Original E-Chart Image



E W S X

E W S X M

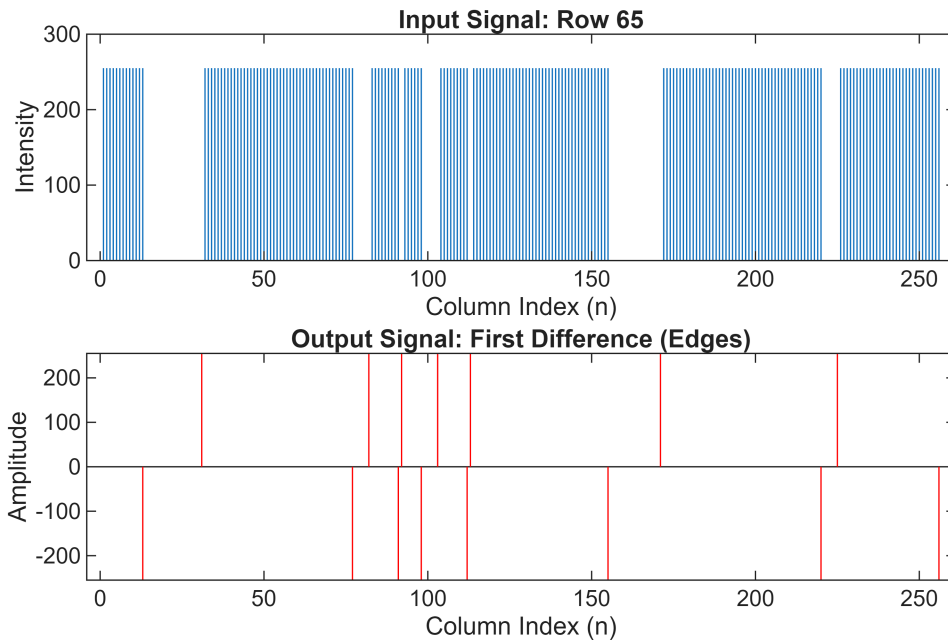
E W S X M P

```
% This treats one line of the image as a 1D signal
row_idx = 65;
x = echart(row_idx, :);

% Apply the first difference filter
h = [1, -1];
y = conv(double(x), h, 'same');

% Visualization: Compare Input vs. Output
figure;
subplot(2,1,1);
stem(x, 'Marker', 'none');
title(['Input Signal: Row ', num2str(row_idx)]);
xlabel('Column Index (n)'); ylabel('Intensity');

subplot(2,1,2);
stem(y, 'Marker', 'none', 'Color', 'r');
title('Output Signal: First Difference (Edges)');
xlabel('Column Index (n)'); ylabel('Amplitude');
```



What do the 0s and 255s represent in the input?

In the grayscale echart image, 0 represents black pixels and 255 represents white pixels.

What do the non-zero values represent in the output $y[n]$?

The non-zero values represent edges. A non-zero value occurs only when there is a change in intensity between adjacent pixels (e.g., switching from the white background to a black letter)

%% Task 4: Bar Code Detection

% Read the image and extract a middle row

```
img = imread('C:\Users\Administrator\OneDrive - Habib University\Sem 4\Signals & Systems\MatLab\Lab07\HP110v3.png');
```

```
[rows, cols, ~] = size(img);
```

```
mid_row = round(rows/2); % Get the middle row index
```

```
x = double(img(mid_row, :));
```

% Applying first Difference filter

```
h = [1, -1];
```

```
y = conv(x, h, 'same');
```

%Thresholding System

```
tau = 100;
```

```
d = abs(y) >= tau; % d[n] is 1 at edges, 0 otherwise
```

% Finding edge locations

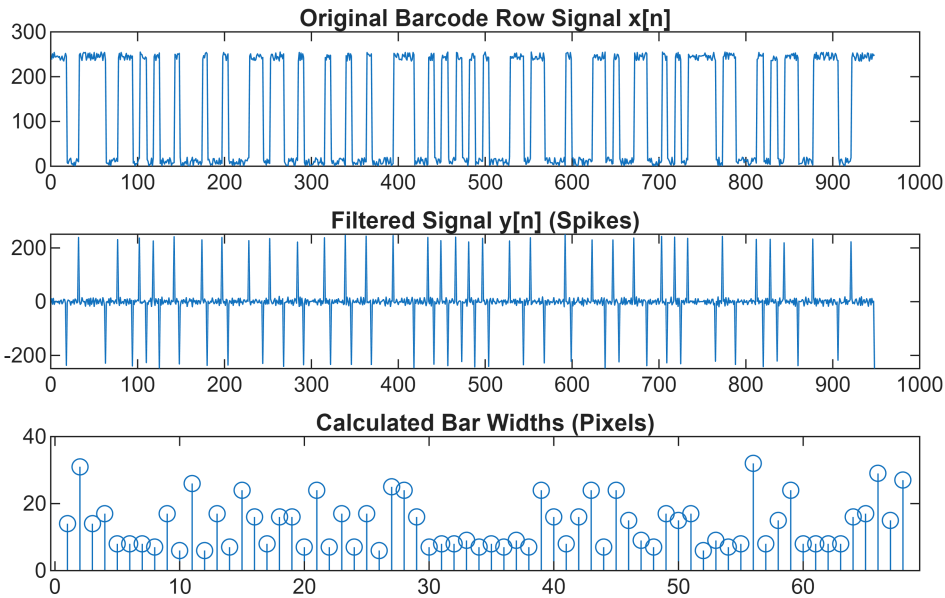
```
locs = find(d);
```

```

% Calculating bar widths
% Difference between consecutive edge locations gives the width of bars
bar_widths = conv(locs, h, 'same');
bar_widths = diff(locs);

% Plotting
figure;
subplot(3,1,1); plot(x); title('Original Barcode Row Signal x[n]');
subplot(3,1,2); plot(y); title('Filtered Signal y[n] (Spikes)');
subplot(3,1,3); stem(bar_widths); title('Calculated Bar Widths (Pixels)');

```



```

%% Task 5: Final Decoding
myPath = 'C:\Users\Administrator\OneDrive - Habib University\Sem 4\Signals & Systems\MatLab\Lab07';
addpath(myPath);

% Extract the 59 bars starting from the 1-1-1 guard pattern
useful_widths = bar_widths(5:63);

% Normalize to the 95-unit standard
w_bar = sum(useful_widths) / 95;
delta_fixed = round(useful_widths / w_bar);

% Force the sum to 95
% If the sum is 96, we will find the largest bar and shrink it by 1.
% If the sum is 94, we find the largest bar and grow it by 1.
while sum(delta_fixed) > 95

```

```

    [~, idx] = max(delta_fixed);
    delta_fixed(idx) = delta_fixed(idx) - 1;
end
while sum(delta_fixed) < 95
    [~, idx] = max(delta_fixed);
    delta_fixed(idx) = delta_fixed(idx) + 1;
end

% 4. Final Decoding
barcode_result = decodeUPC(delta_fixed);

```

```

ssbeg = 1×3
      1      1      1

```

```

disp('The decoded barcode is:');

```

The decoded barcode is:

```

disp(barcode_result);

```

```

      8      8      2      7      8      0      4      5      0      1      6      5

```

RUBRIC:

Assessment Rubric
Lab 7: Convolution-II

Name:	Student ID:
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Marks distribution:

	Tasks	LR2 Code	LR5 Results	LR9 Report	AR4 Report Submission	AR7 Comments
In-Lab	Task 1		/8	/04	/02	
	Task 2	/8	/4			/2
	Task 3	/8	/4			/2
	Task 4	/12	/12			/4
Total =70		/28	/28	/04	/02	/8
CLO Mapped		CLO2	CLO2	CLO2	CLO3	CLO3

CLO – (LDL)	Total Points	Points Obtained
CLO2 – (Psy-4)	56	
CLO3 – (Aff-3)	14	
Total	70	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.